

Assignment 4: Distribution Analysis (Seaborn) The Dataset: Use the built-in 'tips' dataset `tips = sns.load_dataset('tips')` The Question: 1. Create a Histogram (using `displot`) for the `total_bill` column. Include a KDE (Kernel Density Estimate) line and change the color to "darkblue". 2. Generate a JointPlot to compare `total_bill` (x-axis) and `tip` (y-axis). Add a regression line to see the correlation. 3. Create a BoxPlot comparing the `total_bill` across different `days` of the week, colored (hue) by whether the customer was a `smoker`.

Assignment 4: Distribution Analysis using Seaborn

Import required libraries

`import seaborn as sns`

`import matplotlib.pyplot as plt`

`import pandas as pd`

Create Dataset Manually

`data = {`

`'total_bill': [16.99, 10.34, 21.01, 23.68, 24.59,`
 `25.29, 8.77, 26.88, 15.04, 14.78],`

`'tip': [1.01, 1.66, 3.50, 3.31, 3.61,`
 `4.71, 2.00, 3.12, 1.96, 3.23],`

`'day': ['Sun', 'Sun', 'Sun', 'Sun', 'Sun',`
 `'Sat', 'Sat', 'Sat', 'Fri', 'Fri'],`

```

'smoker': ['No', 'No', 'No', 'No', 'No',
           'Yes', 'Yes', 'Yes', 'No', 'Yes']
}

tips = pd.DataFrame(data)

# -----
# 1. Histogram using displot with KDE line
# -----

sns.displot(
    data=tips,
    x='total_bill',
    kde=True,
    color='darkblue'
)

plt.title("Histogram of Total Bill")
plt.xlabel("Total Bill")
plt.ylabel("Frequency")

plt.show()

# -----
# 2. JointPlot with Regression Line

```

```
# -----
```

```
sns.jointplot(  
    data=tips,  
    x='total_bill',  
    y='tip',  
    kind='reg'  
)
```

```
plt.show()
```

```
# -----
```

```
# 3. BoxPlot comparing total_bill across days
```

```
# -----
```

```
plt.figure(figsize=(8,5))
```

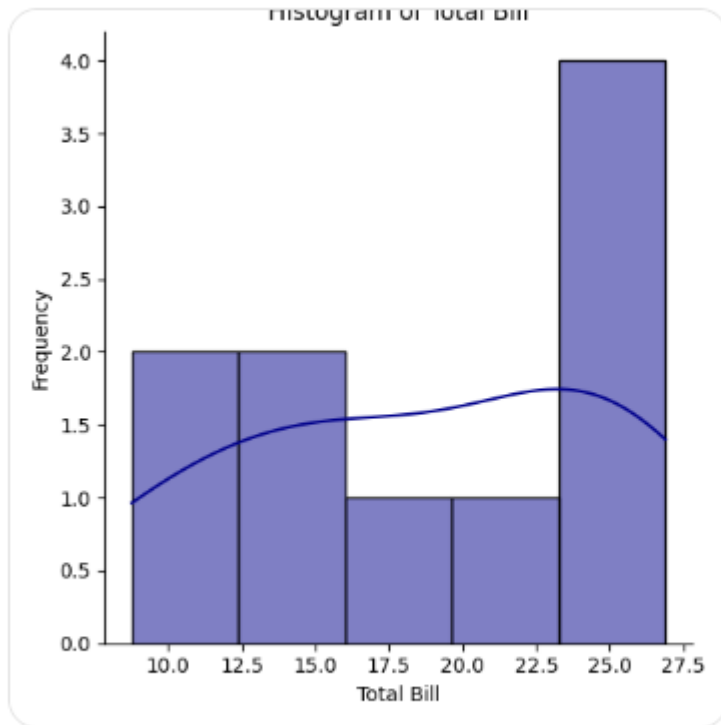
```
sns.boxplot(  
    data=tips,  
    x='day',  
    y='total_bill',  
    hue='smoker'  
)
```

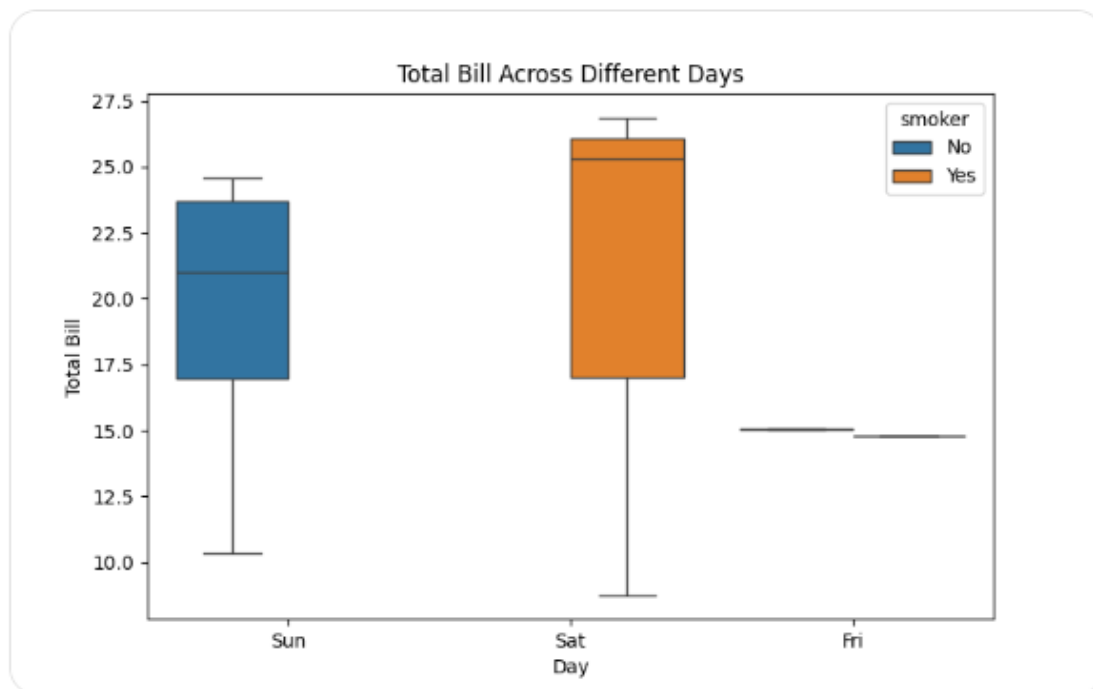
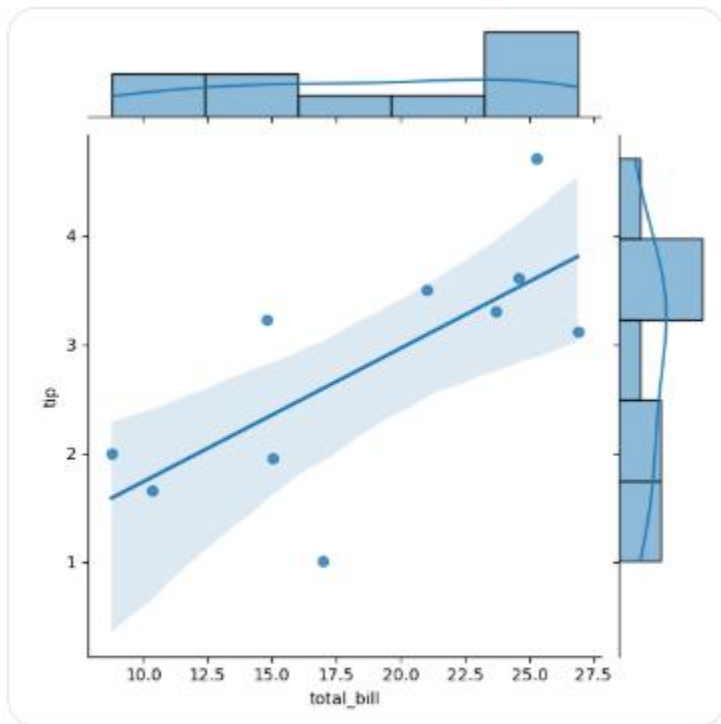
```
plt.title("Total Bill Across Different Days")
```

```
plt.xlabel("Day")
```

```
plt.ylabel("Total Bill")
```

```
plt.show()
```





Assignment 5: Multi-Visual Comparison (Matplotlib) The Dataset: cities = ['Chennai', 'Mumbai', 'Delhi', 'Bangalore', 'Hyderabad'] sales_values = [450, 600, 520, 780, 410] categories = ['Electronics', 'Fashion', 'Groceries', 'Home Decor'] market_share = [35, 25, 30, 10] The Question: Create a single figure with two subplots (1 row, 2 columns): 1. Subplot 1 (Bar Chart): Plot `cities` vs `sales\_values`. Set the bar color to 'teal', add a grid on the Y-axis, and label the axes. 2. Subplot 2 (Pie Chart): Plot the `market\_share` for the four `categories`. Use the `explode` parameter to pop out the "Electronics" slice and show the percentage autoformatted to \$1\%\$. 3. Add a Main Suptitle to the entire figure: "Regional Sales & Category Analysis".

Assignment 5: Multi-Visual Comparison using Matplotlib

Import required library

```
import matplotlib.pyplot as plt
```

Dataset

```
cities = ['Chennai', 'Mumbai', 'Delhi', 'Bangalore', 'Hyderabad']
```

```
sales_values = [450, 600, 520, 780, 410]
```

```
categories = ['Electronics', 'Fashion', 'Groceries', 'Home Decor']
```

```
market_share = [35, 25, 30, 10]
```

```
# -----
```

```
# Create Figure with 1 Row and 2 Columns
```

```
# -----
```

```
fig, axes = plt.subplots(1, 2, figsize=(12, 5))
```

```
# -----  
  
# Subplot 1 : Bar Chart  
  
# -----  
  
axes[0].bar(cities, sales_values, color='teal')  
  
axes[0].set_title("City Sales Comparison")  
axes[0].set_xlabel("Cities")  
axes[0].set_ylabel("Sales Values")  
  
# Add grid on Y-axis  
axes[0].grid(axis='y')  
  
# -----  
  
# Subplot 2 : Pie Chart  
  
# -----  
  
explode = [0.1, 0, 0, 0]  
  
axes[1].pie(  
    market_share,  
    labels=categories,  
    explode=explode,  
    autopct='%1.1f%%'  
)
```

```
axes[1].set_title("Market Share by Category")
```

```
# -----
```

```
# Main Title
```

```
# -----
```

```
fig.suptitle("Regional Sales & Category Analysis", fontsize=16)
```

```
# Adjust layout
```

```
plt.tight_layout()
```

```
# Show output
```

```
plt.show()
```

